



Workshop Daily Planning Board

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC Management

Workshop Daily Planning Board0

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1.0 Introduction

Borusan Makina Kazakhstan's Component Rebuild Shop (CRS) has implemented a daily planning board to efficiently follow up its daily operations.

Due to this board, the workshop supervisor and service engineers can visually see the current volume of work in the workshop. In addition, the board allows the work-in-process starting from Awaiting Disassembly to Component ready to dispatch to be observed.

2.0 Best Practice Description

There are several participants in the component repair process. There is a Shop supervisor, Team leader, Planning Engineer, Spare Parts specialist, and Machine Shop Engineer. Due to the large number of participants in the repair process, important information has the potential to be lost during communication. It influences the turnaround time of the repaired components. The components that were not ready by the planned time may cause complaints from the customers. Besides, the planning accuracy of the shop was far from the target. That is why an idea to represent visually the work-in-process (WIP) of the shop has been implemented.

The board structure is quite simple. It consists of Engine and PT work orders, which go through the repair process on the board. At our dealership, the planning engineer is responsible for constantly updating the workflow board.

Clarifications to the Board:

- **Awaiting Disassembly** – the work orders awaiting disassembly are put on this section;
- **Disassembly** – the work orders being disassembled are put on this section;
- **Defect list preparation** – the work orders for which defect lists are prepared;
- **Technical Report preparation** – the work orders for which technical reports are prepared;
- **Awaiting Defect list from MS (Machine Shop)** – when we are awaiting information on salvage work.
- **Awaiting answer from CAT** – when we need information on components from the CAT.
- **Awaiting repair** – the work orders approved by Customers and put on the plan are shown here, but the shop have not touched them yet due to several reasons (dates of spare parts arrival and MS salvage are not known or their dates are known but it will take long time until the spare parts come and salvage work is done).
- **Awaiting spare parts** – the work orders awaiting spare parts are shown on this section. MS salvage is finished;
- **Awaiting salvage** – the work orders with 100% of spare parts awaiting salvage are shown here. Usually salvage goes parallel with awaiting spare parts, but sometimes salvage is done after all spare parts are delivered. In this case we have to wait until salvage is finished and after that assemble the component;
- **Awaiting spare Parts, partial assembly** – the work orders where we can start assembling subcomponents of the component until we are waiting for the rest spare parts.
- **Awaiting Assembly** – the work orders with 100% of spare parts but cannot be assembled due to the lack of people or spare bays;
- **Assembly** – the work orders being assembled are put on this section.

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- **Awaiting Test** – the work orders assembled but cannot be tested due to the lack of people or spare bays;
Test – the work orders being tested are put on this section.
Additional awaiting – the work orders that are finished but cannot be dispatched. For example, the finished torque converter is awaiting the completion of the engine overhaul. As soon as the engine is finished, they will be connected and then dispatched to the Customer.
- **COMPLETED** (*components are ready*) – the work orders finished are shown here;
- **COMPLETED** (*Defect list and Technical report are sent to be quoted*) – the work orders on which defect lists and technical reports have been prepared and sent to be quoted, and now they are waiting for Customer's approval.

Awaiting Disassembly	Disassembly	Defect List (DL) Preparation	Awaiting Repair	Awaiting spare parts	Awaiting Salvage	Awaiting spare parts, Partial assembly	Awaiting Assembly	Awaiting Test	COMPLETED (<i>components are ready</i>)
		Technical Report Preparation							
		Awaiting DL from Machine Shop					Assembly	Test	Completed (<i>Defect List and Technical report are sent to be quoted</i>)
		Awaiting answer from CAT						Additional waiting	
ENGINES									
Awaiting Disassembly	Disassembly	Defect List (DL) Preparation	Awaiting Repair	Awaiting spare parts	Awaiting Salvage	Awaiting spare parts, Partial assembly	Awaiting Assembly	Awaiting Test	COMPLETED (<i>components are ready</i>)
		Technical Report Preparation							
		Awaiting DL from Machine Shop					Assembly	Test	Completed (<i>Defect List and Technical report are sent to be quoted</i>)
		Awaiting answer from CAT						Additional waiting	
PT components									

Picture 1. Our Workshop Daily Planning Board structure

Clarifications to the magnetic stripe template that is used in the Board:

- **Work Order No.** – the work order number is written here;
- **Customer Name** – the customer name is written here;
- **Model** – the engine or machine model is written here;
- **Component** – the component name is written here;
- **Spare Parts, Start Assy Date** – the date of component assembly starting is written here;
- **Last Spare Part Coming Date** – the date when the last spare part will come is written here;

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- **MS, Start Salvage Date** – the date when the Machine Shop is planning to start salvaging of parts;
- **MS, Last Part Date** – the date when the last salvaged part comes from the Machine Shop is written here;
- **PTD Plan Date** – the date when the Planning-and-Technical Department puts this work order in the plan, by this date the Component Rebuild Shop (CRS) must finish the work order (component);
- **CRS Plan Date** – the date when the CRS is planning to finish this work order (component) as per its availability.

Work Order No.	Model	Spare Parts, Start Assy Date	MS, Start Salvage Date	Disassy Start Date	PTD Plan Date
Customer Name	Component	Last Spare Part Coming Date	MS, Last Part Date	Disassy Finish Date	CRS Plan Date

Picture 2. Our magnetic stripe template (for the Board)

3.0 Implementation Steps

In order to implement the Workshop Daily Planning Board in daily operations:

- 1) Purchase the equipment required (metallic board, magnetic stripes, markers);
- 2) Think over what information will be on a magnetic stripe and order the quantity you required;
- 3) Transfer the repair process and downtime into the board (awaiting disassembly, disassembly, awaiting parts, etc.);
- 4) Start using the board for your daily WIP components. For example, we (Shop supervisor, Team leader, Planning Engineer, Spare Parts specialist, and Machine Shop Engineer) makes meetings beside this Board 3 times a week and discuss the current status, plans, concerns, etc. This Board visually helps to see the WIP and not to miss anything during the repair process.

See below the photo of Workshop Daily Planning Board implemented in our daily operations.



Picture 3. Our Workshop Daily Planning Board in use

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4.0 Benefits

- Visual representation of WIP components on a daily base;
- Easy to use and follow up WIP components;
- Improved planning of work-in-process components;
- Improved communication between shops and related departments;
- Correct components repair history data;
- Increased Customers' loyalty to the CRC.

5.0 Resources Required

- 3-meter metallic board;
- Magnetic stripes with the information needed to stick on the board;
- Markers (that can be erased from the magnetic stripe when the work order is finished).

6.0 Supporting Attachments / References

N/a.

7.0 Related Best Practices

N/a.

8.0 Acknowledgements

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